

Programme: B.E
CIE: Test 1
Credits: 2 : 1 : 0
Sem: IV

Department of Artificial Intelligence and Machine Learning
Term: May-August 2023
Subject: Theory of Computations

Max Marks: 30
Subject Code: A144

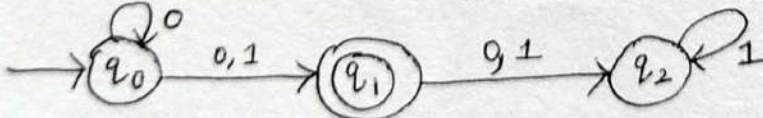
Date: 12/06/2023

Time: 9:30 -10:30 am

Portions for Test: (L1-L15)

Instructions to Candidates:

1. Question 1 is Compulsory. Answer any one question from 2 and 3.
2. Each Question carries 15M. Write figures wherever necessary.
3. Mobiles/Smart Gadgets are restricted.

Sl#	Question	Marks	Bloom's Level	CO Mapping																										
1	a. Design DFAs to accept the following languages. i. String beginning with 'bab' over $\Sigma = \{a,b\}$. ii. String not ending with 011 over $\Sigma = \{0,1\}$.	05	Apply	CO2																										
	b. Convert the following NFA to a DFA over $\Sigma = \{a,b\}$. <table border="1"><thead><tr><th>δ NFA</th><th>ϵ</th><th>a</th><th>b</th></tr></thead><tbody><tr><td>$\rightarrow Q0$</td><td>$\varnothing$</td><td>$\varnothing$</td><td>Q1</td></tr><tr><td>* Q1</td><td>Q2</td><td>Q1</td><td>Q2</td></tr><tr><td>Q2</td><td>$\varnothing$</td><td>$\varnothing$</td><td>Q0</td></tr></tbody></table>	δ NFA	ϵ	a	b	$\rightarrow Q0$	$\varnothing$	$\varnothing$	Q1	* Q1	Q2	Q1	Q2	Q2	$\varnothing$	$\varnothing$	Q0	05	Apply	CO3										
	δ NFA	ϵ	a	b																										
$\rightarrow Q0$	$\varnothing$	$\varnothing$	Q1																											
* Q1	Q2	Q1	Q2																											
Q2	$\varnothing$	$\varnothing$	Q0																											
c. Consider the DFA. <table border="1"><thead><tr><th></th><th>0</th><th>1</th></tr></thead><tbody><tr><td>$\rightarrow A$</td><td>B</td><td>A</td></tr><tr><td>B</td><td>A</td><td>C</td></tr><tr><td>C</td><td>D</td><td>B</td></tr><tr><td>*D</td><td>D</td><td>A</td></tr><tr><td>E</td><td>D</td><td>F</td></tr><tr><td>F</td><td>G</td><td>E</td></tr><tr><td>G</td><td>F</td><td>G</td></tr><tr><td>H</td><td>C</td><td>D</td></tr></tbody></table> i. Identify all distinguishable and indistinguishable pairs. ii. Construct the minimum state equivalent DFA.		0	1	$\rightarrow A$	B	A	B	A	C	C	D	B	*D	D	A	E	D	F	F	G	E	G	F	G	H	C	D	05	Apply	CO3
	0	1																												
$\rightarrow A$	B	A																												
B	A	C																												
C	D	B																												
*D	D	A																												
E	D	F																												
F	G	E																												
G	F	G																												
H	C	D																												
2	a. Differentiate between DFA, NFA and epsilon- NFA	05	Analyze	CO3																										
	b. Show that the language L consisting of all palindromes over $(0+1)^*$ is not regular	05	Apply	CO3																										
	c. Give DFA accepting the following language for the alphabet 0,1. Such that: the set of all strings beginning with a '1' that intercepted as binary integer is a multiple of 5.	05	Apply	CO2																										
3	a. State and prove pumping lemma for regular languages.	05	Understand	CO3																										
	b. Convert the following NFA into an equivalent DFA. 	05	Apply	CO2																										
	c. Define the following with an example each. a) Language of DFA. b) Language of NFA. c) Power of an Alphabet. d) If M is a DFSM and $\epsilon \in L(M)$, what simple property must be true of M? e) How Many Start and Final / Accepting a Finite Automata can have?	05	Remember	CO1																										

Course Outcomes meant to be assessed by the IA Test 1:

CO1: Explain the basic concepts of formal languages and finite automata. (PO-1, PSO-1)

CO2: Construct automata to accept strings from a specified language. (PO-1,2,3, PSO-1)

CO3: Convert among equivalently powerful notations for a language, including among DFAs, NFAs, and regular expressions, between PDAs, CFGs and normal forms of CFGs. (PO-1,2,3, PSO-1)

Internal Assessment Question Paper – 2

Ramaiah Institute of Technology

(Autonomous Institute, Affiliated to VTU)

Department of Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning

Programme: B.E

Term: May-August 2023

Max Marks: 30

CIE: Test 2

Subject: Theory of Computations

Subject Code: AI44/AD44

Credits: 2 : 1 : 0

Sem: IV

Date: 27/07/2023

Time: 9:30 -10:30

Portions for Test: (L16-

L28)

Instructions to Candidates:

1. Question 1 is Compulsory. Answer any one question from 2 and 3.
2. Each Question carries 15M. Write figures wherever necessary.
3. Mobiles/Smart Gadgets are restricted.

Sl#	Question	Marks	Bloom's Level	CO Mapping															
1	a. Design PDA of the Language $L = \{ww^R \mid w \text{ is in } (0+1)^*\}$	05	Apply	CO 3															
	b. Begin with the grammar: $S \rightarrow ABC \mid BaB$ $A \rightarrow aA \mid BaC \mid aaa$ $B \rightarrow bBb \mid a \mid D$ $C \rightarrow CA \mid AC$ $D \rightarrow \epsilon$ i. Eliminate ϵ – productions. ii. Eliminate any unit productions in the resulting grammar. iii. Eliminate any useless symbols in the resulting grammar. iv. Put the resulting grammar into Chomsky Normal Form.	10	Apply	CO 4															
2a	i. Design CFG for the language of palindromes over $\{0,1\}$. ii. Design CFG over $\{a, b\}$ consisting of all strings of even length. iii. Abbreviate and define CFG.	05	Analyze	CO 3															
b	State and prove pumping lemma for CFL.	05	Apply	CO 4															
c	Design Turing Machine for the following language $L = \{a^n b^n c^n, n \geq 1\}$, Also draw Transition diagram for the TM.	05	Apply	CO 5															
3a	Given an instance of PCP, tell whether this instance has a solution. <table border="1" data-bbox="253 1364 595 1543"> <thead> <tr> <th></th><th>List A</th><th>List B</th></tr> </thead> <tbody> <tr> <td>i</td><td>w_i</td><td>x_i</td></tr> <tr> <td>1</td><td>110</td><td>110110</td></tr> <tr> <td>2</td><td>0011</td><td>00</td></tr> <tr> <td>3</td><td>0110</td><td>110</td></tr> </tbody> </table>		List A	List B	i	w_i	x_i	1	110	110110	2	0011	00	3	0110	110	05	Understand	CO 5
	List A	List B																	
i	w_i	x_i																	
1	110	110110																	
2	0011	00																	
3	0110	110																	
b	Consider the grammar $S \rightarrow SbS \mid a$ This grammar is ambiguous; show in particular that the string abababa has two. i. Parse trees ii. Leftmost derivations iii. Rightmost derivations.	05	Apply	CO 3															
c	Define Turing Machine with neat diagram of its abstract Structure. What is the language accepted by Turing machine?	05	Remember	CO 5															

Course Outcomes meant to be assessed by the IA Test 2:

CO3: Convert among equivalently powerful notations for a language, CFGs and Normal forms of CFGs.

CO4: Prove the various closure and decision properties of formal languages.

CO5: Explain the concepts of Turing Machines, Post Correspondence problem.